



CONTACT

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 Driver's License

LANGUAGES

 French - Native

 English - Intermediate (B1 / B2)

 Spanish - Elementary (A2 / B1)

SOFT SKILLS

- Rigor & Critical Thinking
- Self-Starter & Curiosity
- Discipline & Perseverance

INTERESTS

 Astronomy

 Running

 Esports

 Cinema

Bastien BABU

STUDENT IN DOUBLE BACHELOR'S DEGREE IN MATHEMATICS & COMPUTER SCIENCE

EDUCATION

2023-2026

Double Bachelor's Degree in Mathematics & Computer Science

University of Toulouse - Paul Sabatier

- Mathematics : Algebra, Analysis, Probability, Statistics
- Computer Science : Algorithms, Machine Learning, Software Engineering

2023

High School Diploma - Highest Honors

Lycée Notre Dame, Challans

- Major : Mathematics, Computer Science (NSI), English (AMC)
- Elective : Advanced Mathematics

EXPERIENCE

Jan 2026 - Present

Research Internship - AI & Medical Imaging

Supervised by Duong Hung PHAM - MINDS Team, IRIT Laboratory, Toulouse

Generation of synthetic ultrasound images using diffusion models (DDPM) for medical image segmentation.

- Literature review and reproduction study of "Echo from noise" (Stojanovski et al., 2023)
- Synthetic data generation and training of segmentation neural networks
- Experimental evaluation on simulated and real-world datasets

Oct 2025 - Jan 2026

AI Project - Computer Vision & Biomechanics

Kinematic analysis and human pose estimation applied to tennis.

- Detection of 17 key body points using HRNet (MMPose / OpenMMLab).
- Kinematic computations (velocity, acceleration, joint angles) and 2D visualization (OpenCV, Matplotlib).
- Trajectory prediction using LSTM recurrent neural networks (TensorFlow)

GitHub : [ByNeex/Tennis-Pose-Analysis](#)

Jan 2025 - Jun 2025

Research Project - Mathematical Analysis

Supervised by Patrice Lassère - Institute of Mathematics of Toulouse (IMT)

Exploration of Gamma and Beta functions, theory and applications.

- In-depth theoretical study of Gamma and Beta functions : integrals, infinite products, analytic continuation.
- Applications in probability theory, volume computation of hyperspheres...
- Scientific report written in LaTeX (graded 18/20).

Report available upon request : [Consulter le PDF](#)

TECHNICAL SKILLS

Mathematics : Probability, Statistical Inference, Modeling.

Programming : Python, C, Java, Assembly (ARMv7 32 bits), SQL, OCaml, Coq, R.

Libraries : PyTorch, TensorFlow, Scikit-learn, Pandas, MMPose, OpenCV.

Web : HTML/CSS, JavaScript (personal website).

Tools : Jupyter, Git, VS Code, Google Collab.